During the Spring 2025 Semester, I took the course EE590, Consumer Optoelectronics, taught by Dr. Ping-Chin Cheng. In this course, Dr. Cheng covered the various optical, IR, and other devices he's worked with throughout his career. He has experience in the photolithography and biological sciences field, so his perspective in lectures was very eye-opening. Dr. Cheng covered a variety of topics in lecture, but was unafraid to stray from the per-planned course topics in terms of the content he covered, especially if the content was relevant to a student's questions or current developments in the field. Overall, I am glad that I took this course. Although I am not pursuing a career in the field of photolithography or other optical devices, I learned a lot. Dr. Cheng's teaching was very helpful. As he taught me how to better my analytical thinking as an engineer and make connections between seemingly unrelated topics.

In EE590, the main contributor to our semester grade for the semester was our final project. This project was a group project in which we as students formed groups of up to 5 members and chose a project from a list of topics preselected by Dr. Cheng. I knew that I had to work my hardest on this project because I was working with groupmates who were already my friends. I had been in some difficult situations previously and heard of other situations where groups ruined friendships because the group members did not communicate their expectations clearly. Thankfully, I have experienced the opposite effect in which all of my friendships with my group members have been strengthened as we all worked hard to complete the project.

The project we chose was a blood oximeter project, in which we made a blood oximeter with functionality similar to those on the market currently. This device, the blood oximeter, is a common consumer medical device that uses light to shine through a person's finger to detect their blood oxygen level. The oxygenated blood absorbs the infrared light, while the deoxygenated blood absorbs the red light that shines through when the device is clamped upon a person's finger. The difference in the wavelengths of the red and infrared light allow for the two different blood statuses to be measured. To assenmble this project, we used a variety of electronic components. Shown in Figure 1 is the Pulse Oximeter sensor that records the red and infrared light readings. We used other components to finish building the hardware, and connected it to a mobile application to create readings for mobile phone users. (See Figure 2 for system Diagram).

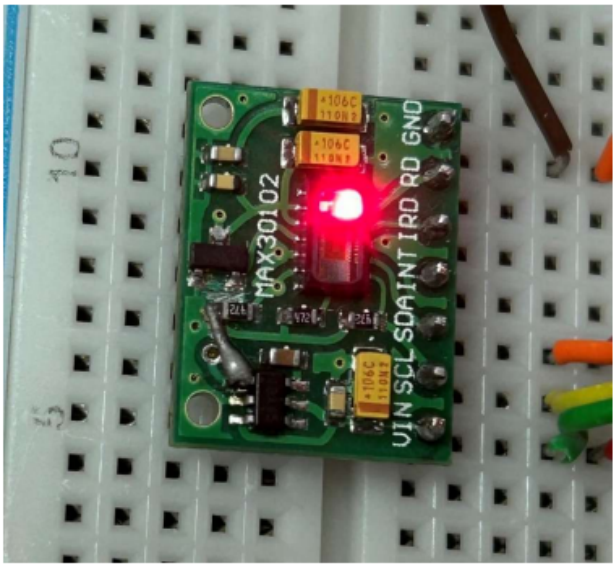

Figure 1: Pulse Oximeter

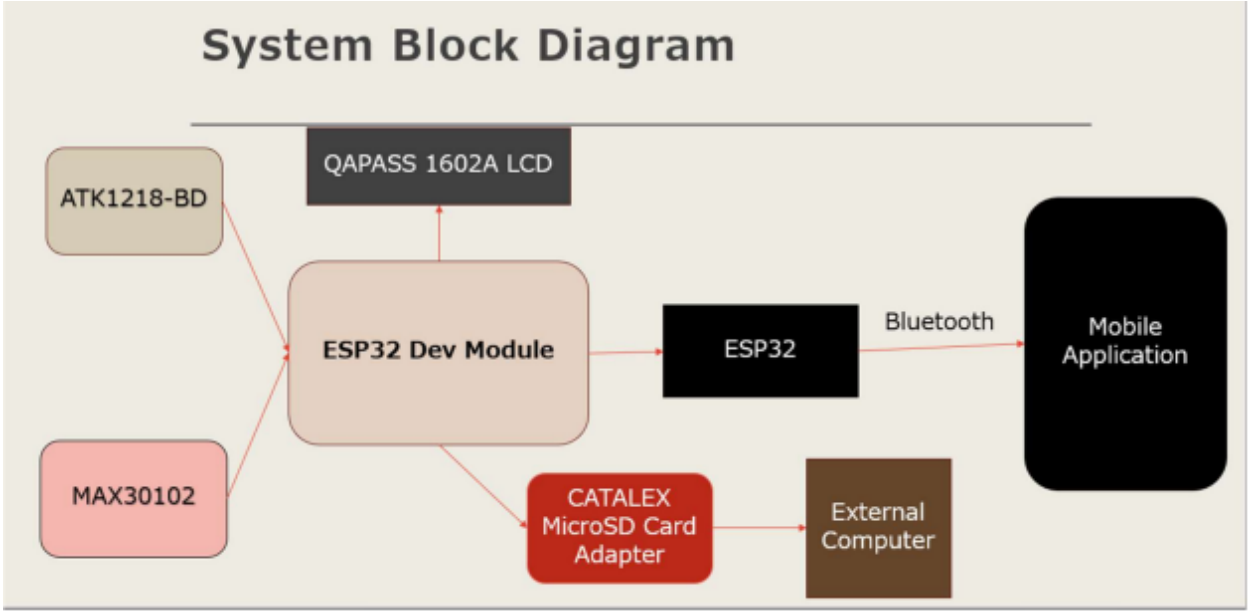

Figure 2: System Diagram

Overall, EE590 was a great class that I have fond memories of. The lectures with Dr. Cheng were very informative yet also lighthearted. The work sessions with my groupmates were productive and fun simultaneously. I am thankful to Dr. Chen for structuring the class in a way that gave us , the students, the freedom to choose which project we work on, and what discussions to pursue in class.